Lecture 8: Estimation

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

AGENDA

01

THE LOGIC OF ESTIMATION

Which parameter is wanted, how an estimator is built, and how a good one is told from a bad one

02

FROM POINT TO INTERVAL

Why a single number is never enough, and the pivotal quantity that turns one into an interval

03

z, t AND χ^2

Three pivots, one method: what each one estimates, what it assumes, and what it costs

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REVISED FROM LECTURE 6

Parameter and Sample Statistic

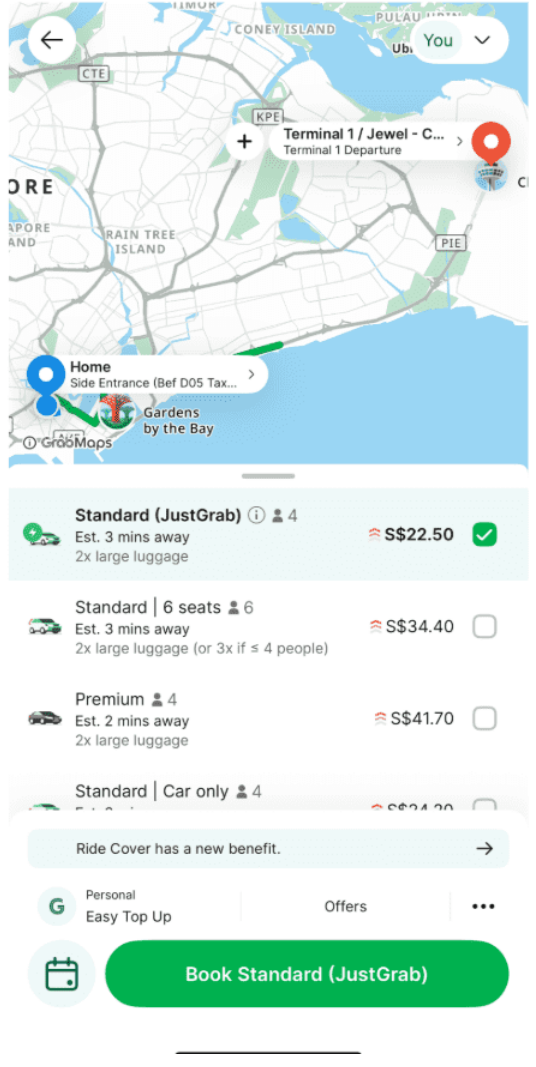
	Parameter (tham số)	Statistic (thống kê mẫu)
Notation	μ, σ, p (mu, sigma, p)	$\bar{x}$, s, $\hat{p}$
Describes	the **Population**	the **Sample**
In practice	unknown	computed
Today, in this room	known — we declared it	the 10 values on the board
Nature of the object	a fixed number	a random variable

THE LOGIC OF ESTIMATION

Why we estimate?







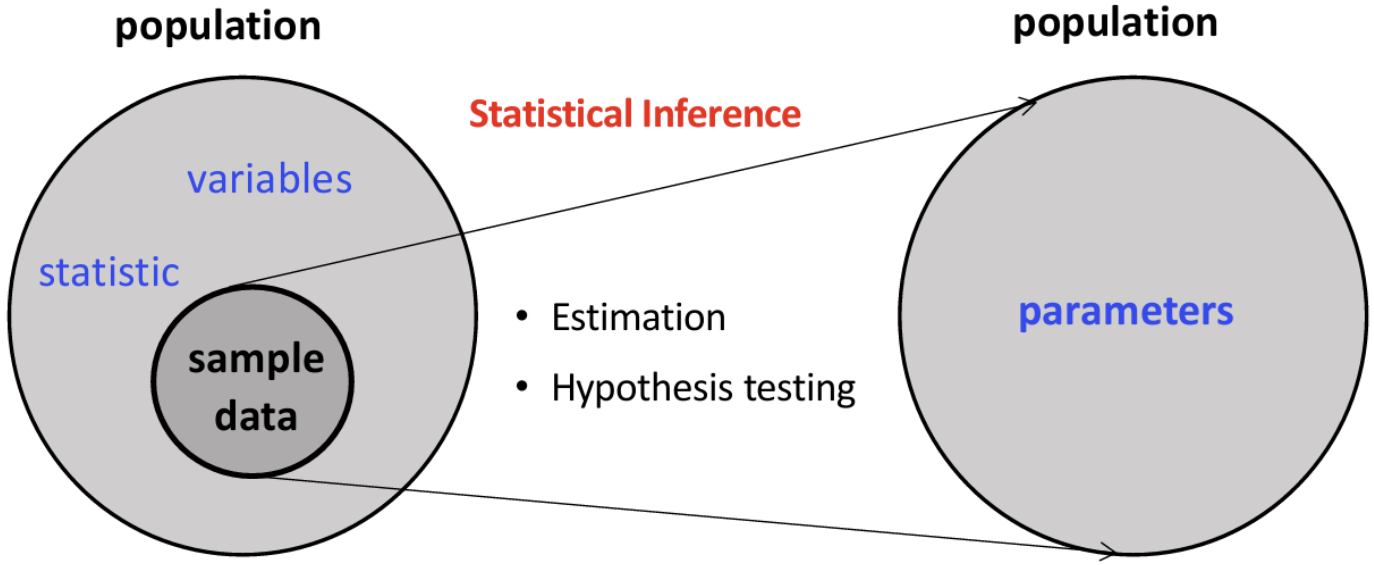




Why we estimate?

Statistical inference is the ways of drawing conclusions about population parameters from an analysis of the sample data

- **Sample** is only a part of the **Population**, it cannot be expected to give us the exact value of the parameter.
- The observed value of a statistic depends on the particular sample selected.
- There will be some **variability** in the value of a statistic over different occasions of sampling.



We wish to make generalizations about a population when only a sample will be available

THE LOGIC OF ESTIMATION

Parameter, Estimator, Estimate

Three distinct objects that ordinary language collapses into one word

Object	Notation	What it is
Parameter (tham số)	θ — here μ, σ^2, p, β	a fixed number belonging to the population; unknown, and it does not vary
Estimator (hàm ước lượng)	$\hat{\theta} = g(X_1, \ldots, X_n)$	a FUNCTION of the sample — hence a random variable, with a distribution of its own
Estimate (giá trị ước lượng)	$\hat{\theta}$	the single number this particular sample produced; one realisation of θ

Almost every confusion in inference is these three collapsed into the phrase "the mean".

Which Parameter, and What the Number Claims

Estimation begins by choosing WHAT is to be estimated — not by choosing a formula

Parameter	Estimator	What the number claims	What it does not claim
μ — population mean	$\bar{x}$	the average taken over every firm in the population	not the typical firm, when the variable is skewed
σ^2 — population variance	s^2	how widely FIRMS differ from one another	not how precisely $\bar{x}$ is known
p — population proportion	$\hat{p}$	the share of the population in that category	not the probability for one named firm
β — population slope	b_1	the ROA gap between two firms one unit apart in X	not the change in one firm over time

For a right-skewed variable the mean and the median answer different questions. Choosing between them is not a technical choice.

THE LOGIC OF ESTIMATION

Type of Estimators

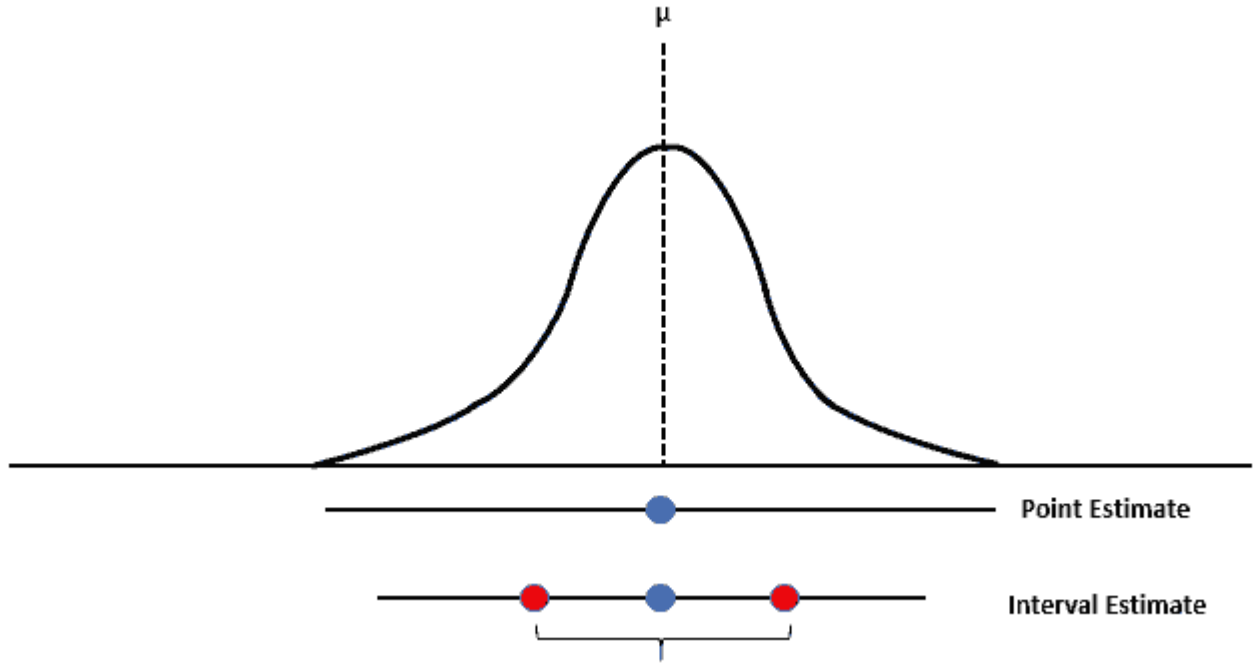
Estimators are not given. Each is produced by a principle, and the principle is a choice

Point Estimate

A single number calculated from sample data to estimate a population parameter.

Interval Estimate

A range of values (derived from a Confidence Interval) used to estimate a population parameter.



THE LOGIC OF ESTIMATION

Three Properties of an Estimator

The two terms of the MSE identity, plus what happens as n grows

NAMING THE TERMS

Unbiased · Efficient · Consistent

The first two are the two terms of the identity on the preceding slide. The third is what happens to both as the sample grows.

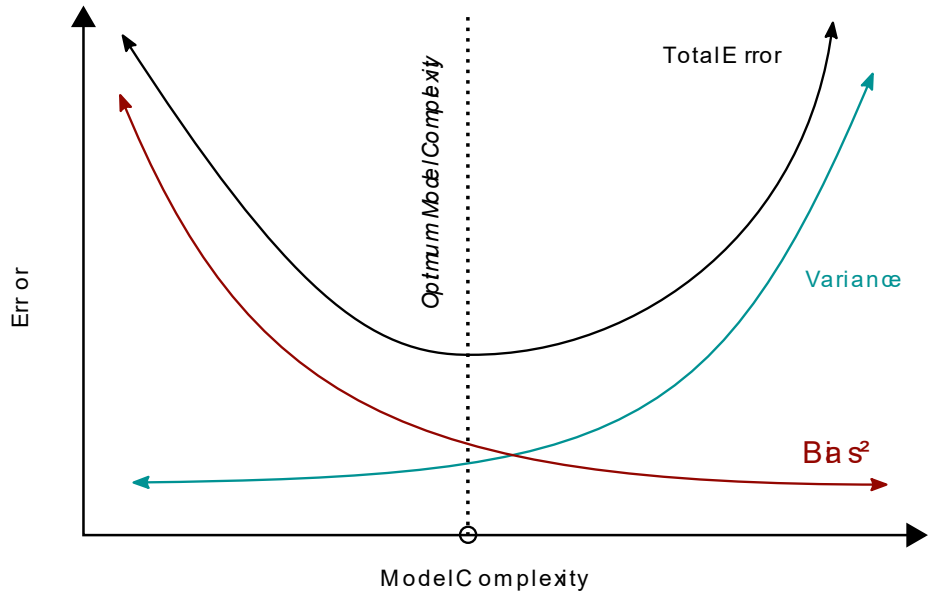
- Unbiased — Bias(θ) = 0: the shots surround the centre
- Efficient — smallest Var among unbiased estimators
- Consistent — $\hat{\theta} \to \theta$ in probability as $n \to \infty$
- MLE is generally consistent and efficient, but not always unbiased

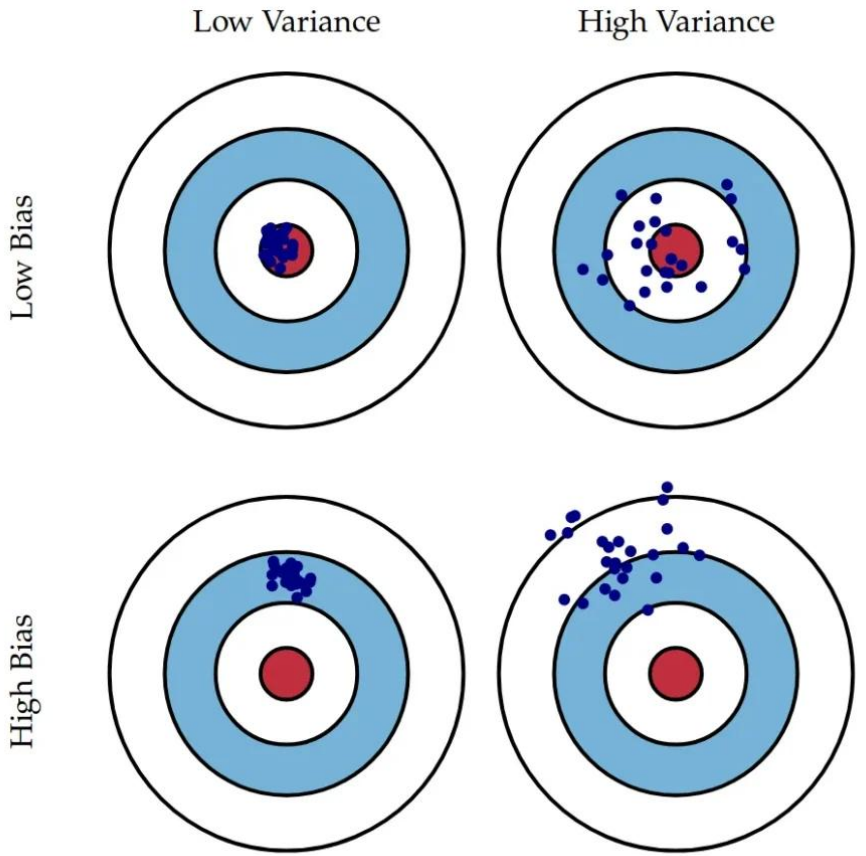
Bias and Variance of an estimator

Bias (độ lệch) is how far the centre of the estimator's distribution sits from the parameter.

Variance (phương sai) is how widely the estimator scatters about its own centre.

The law of Complexity: model cannot be simultaneously rigid **(to keep variance low)** and highly flexible **(to keep bias low)**, they must trade off





Bias, Variance, and Mean Squared Error

Bias is the difference between this estimator's expected value and the true value of the parameter being estimated

Variance is the degree of dispersion (volatility) of the estimators around their own mean value

They both show how “accurate” the estimation is. But they are different terms.

Mean Squared Error (MSE) is the average squared difference between the estimated values and the true value

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

$$\text{Variance}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$$

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \text{Bias}(\hat{\theta})^2 + \text{Variance}(\hat{\theta})$$

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FROM POINT TO INTERVAL

The Direction of the Argument Reversed

From the population downward, to the sample upward

	Lecture 6	Today
Known	μ and σ	one sample — $\bar{x}$ and s
Unknown	which sample will be drawn	μ
Direction	population → sample	sample → population
Object produced	a distribution of $\bar{x}$	an interval for μ

The 2 entities does not change. It is read from the other end.

FROM POINT TO INTERVAL

The Central Region of the Sampling Distribution

Lecture 6, restated on a number line

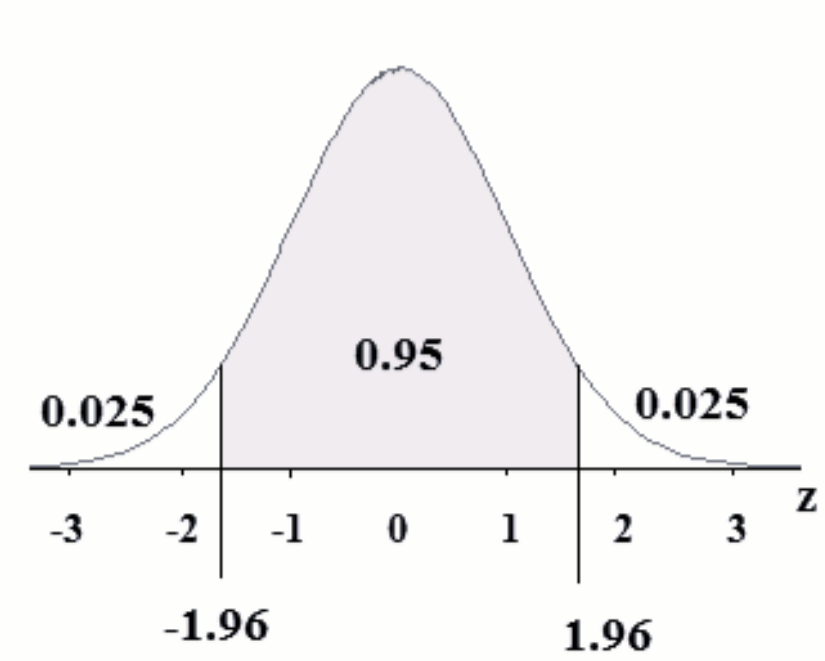
NORMAL DISTRIBUTION

95% of samples land within 1.96·SE of μ

The region from μ − 1.96·SE to μ + 1.96·SE is a property of the population.

It is fixed, but it is not observable, because μ is not known.

- μ — fixed, unknown
- the region around μ — fixed, unknown
- $\bar{x}$ — random: the sample is the thing varies



FROM POINT TO INTERVAL

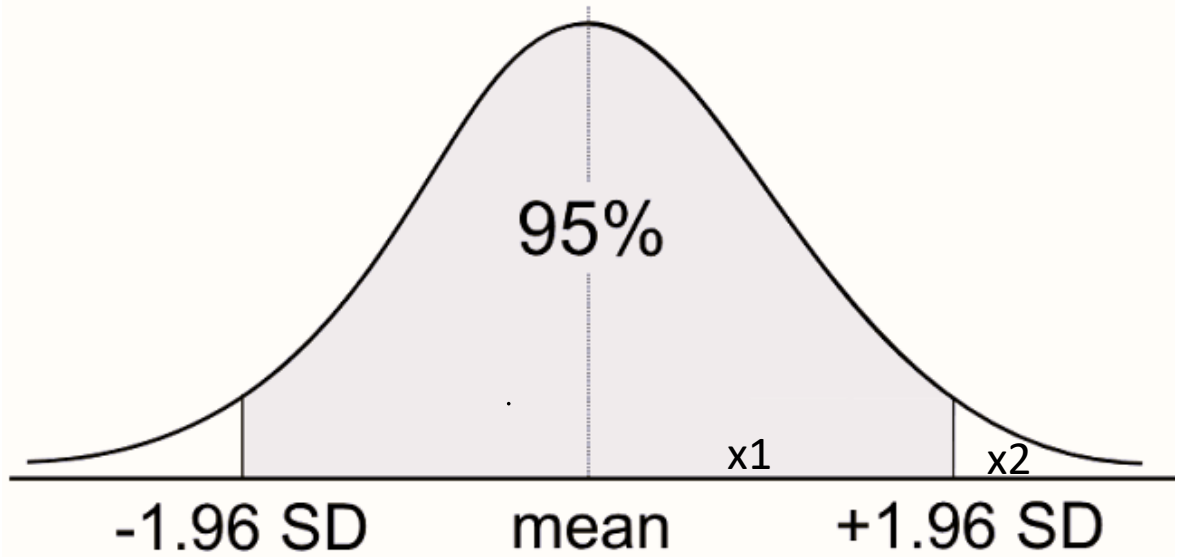
An Interval Built Around a Sample Mean

THE EQUIVALENCE

$\bar{x}$ inside the band ⟺ the interval around $\bar{x}$ covers μ

Take an $\bar{x}$ inside the central band and draw an interval of the same half-width around it: it reaches back to μ. Take an $\bar{x}$ outside the band and draw the same interval: it does not reach μ.

- Inside the band → the interval covers μ
- Outside the band → it fails to cover
- 95% of samples land inside → 95% of intervals cover



The Same Statement in Algebra

$$P(\mu - 1.96 \cdot SE \leq \bar{x} \leq \mu + 1.96 \cdot SE) = P(\bar{x} - 1.96 \cdot SE \leq \mu \leq \bar{x} + 1.96 \cdot SE) = 0.95$$

The second form reads as though μ were varying. It is not. The random quantity is still x̄; rearranging the inequality moved it to the other side without changing what it is. Ninety-five per cent of samples produce an interval that covers μ — the proportion is a property of the sampling procedure.

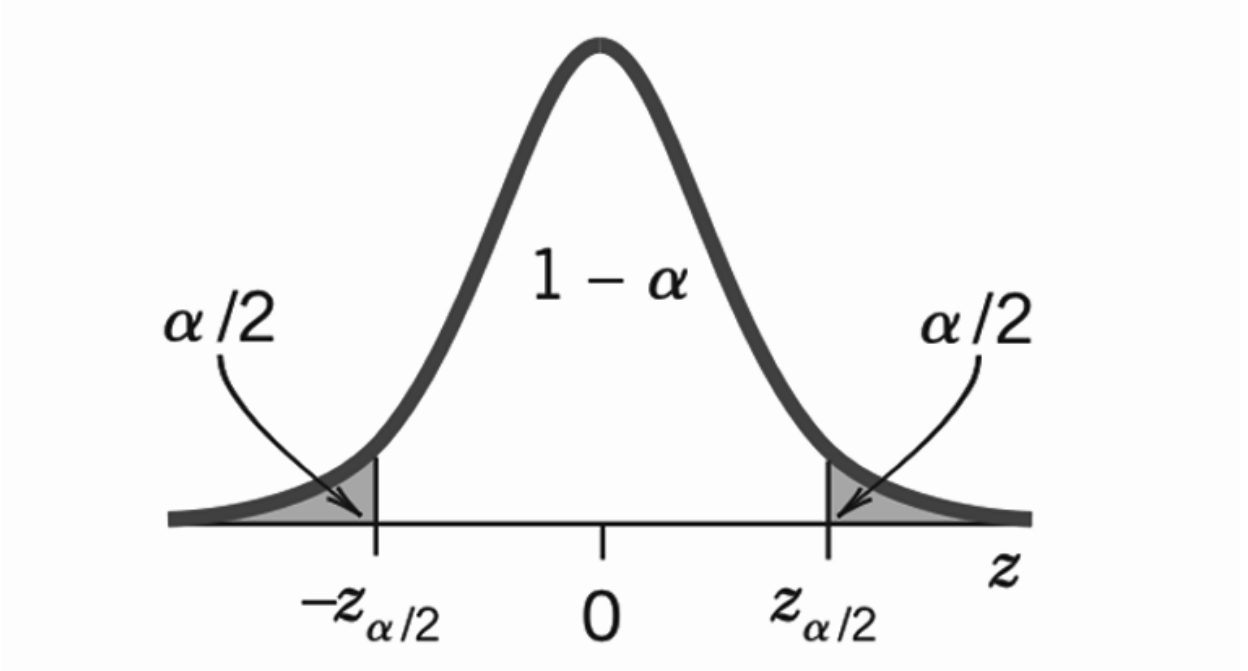
One statement read from two ends. Both intervals have the same length $|\bar{x} - \mu| = |\mu - \bar{x}| \leq 1.96 \cdot SE$

How closely the sample mean is expected to estimate μ?

THE NOTATION

$z_{\alpha/2}$ = the upper α /2 point of the standard normal distribution ($z_{\alpha/2}$ là điểm tới hạn trên ứng với xác suất $\alpha/2$ của phân phối chuẩn).

That is, the area to the right of $z_{\alpha/2}$ is α /2, and the area between - $z_{\alpha/2}$ and $z_{\alpha/2}$ is 1 – α



Some useful values of $z_{\alpha/2}$					
$1-\alpha$	0.80	0.85	0.90	0.95	0.99
$z_{\alpha/2}$	1.28	1.44	1.645	1.96	2.58

Remember $\Phi(z) = P(Z \leq z)$

$$\Phi(z_{\alpha/2}) = 1 - \frac{\alpha}{2}$$

FROM POINT TO INTERVAL

The Anatomy of an Interval Estimate

point estimate ± (critical value) × (standard error) — the inverted pivot, named

01

Point estimate

Ước lượng điểm: the observed value of θ̂, at the centre of the interval.

02

Critical value

Giá trị tới hạn: a quantile of the pivot's distribution: 1.96, or t(n−1), or a χ^2 quantile.

03

Margin of error

Sai số biên: critical value × standard error. Half the width of the interval.

04

Confidence level

Mức tin cậy: $1 - \alpha$, the long-run proportion of intervals that cover θ.

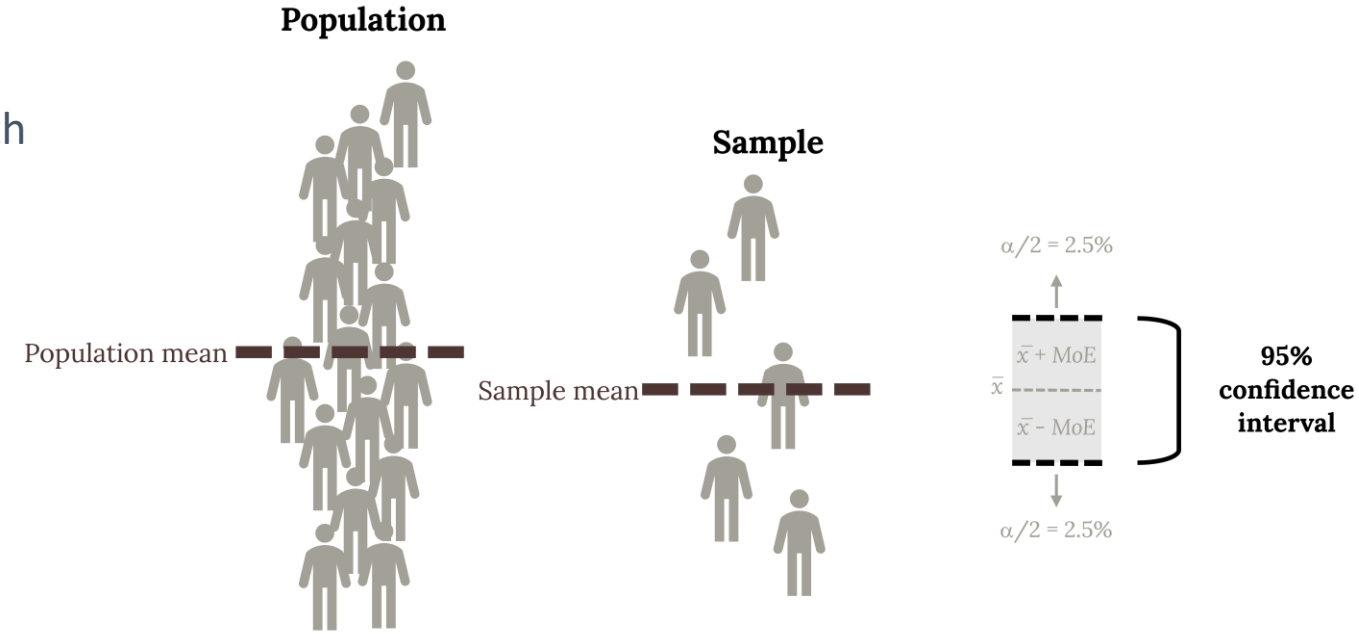
A margin of error reported without its confidence level does not determine an interval.

Confidence Interval for μ

We use a sample to calculate an interval that would definitely contain the true value of the parameter with a specified high probability.

This probability is called the level of confidence is typically taken as 0.90, 0.95, or 0.99

$$\text{CI} = \hat{\theta} \pm z \times \text{SE}(\hat{\theta})$$



THE LOGIC OF ESTIMATION

The Meaning of a Confidence Level

A sample of 40 firms gives a 95% confidence interval for mean ROA: [4.2% , 7.8%]

(A)

95% of firms in the market have an ROA within this interval.

(B)

The probability that the true mean ROA lies within this interval is 95%.

(C)

If sampling were repeated many times and an interval built the same way each time, about 95% of those intervals would contain the true mean ROA.

(D)

There is a 95% probability that the computation was performed correctly.

Recorded now, settled later by the room rather than by me — after the machinery behind the sentence has been built.

The Intervals That Fail

5% of samples produce an interval that does not contain μ, and their authors did nothing wrong.

Those are the samples whose $\bar{x}$ landed in the two tails. The researcher holding such an interval drew correctly, computed correctly and read correctly — and has no way of knowing, because μ is unknown and no diagnostic reports the failure. The 5% is not an imperfection awaiting correction: it is the quantity the method discloses, and it was chosen by the analyst at the moment α was chosen.

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Z, T AND X^2

Three Pivots, One Method

The whole of z, t and χ^2 in one table

Pivot	Distribution	Interval for	When to use
$(\bar{x} - \mu) / (\sigma/\sqrt{n})$	N(0, 1) — the z statistic	μ	σ known, n is large
$(\bar{x} - \mu) / (s/\sqrt{n})$	t(n − 1) — the t student statistic	μ	σ unknown, n is small
$(n - 1)s^2 / \sigma^2$	$\chi^2(n - 1)$ — the Chi-squared statistic	σ^2	Frequency/ categorical
Common structure	numerator: the error of the estimator	denominator: that estimator's own spread	→ unit-free

Three Determinants of the Width

Quantity	Increasing it	Controlled by
n — sample size	narrows the interval, by √n	the researcher, at a cost
s — dispersion of the data	widens the interval	no one
confidence level	widens the interval	the researcher, by choice
Halving the width	requires quadrupling n	the square-root law of Session 6

Z-statistic

When σ is known

A **Z-statistic** measures exactly how many standard deviations a specific data point or sample mean is away from the population mean

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Why n large enough is also equivalent to σ *is known* ?

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10202	0.10027	0.09853
-1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22362	0.22062	0.21770	0.21476
-0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189

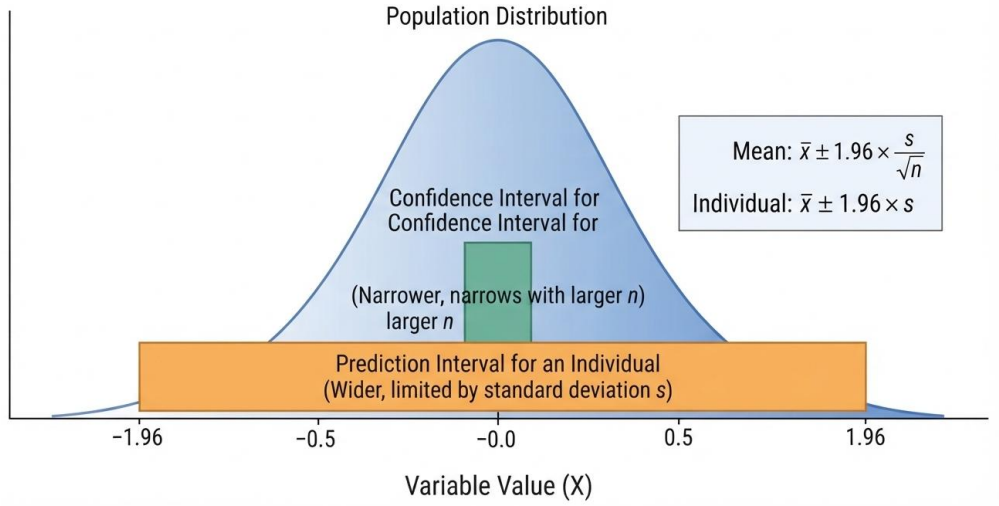
Two Intervals Frequently Confused

DIFFERENT BY A FACTOR OF √N

One describes firms, the other describes a mean

Both are computed from the same sample of 50. The range containing 95% of firms has half-width 1.96·s. The 95% confidence interval for μ has half-width 1.96·s/√n — narrower by about 7 times.

- 95% of firms: $\bar{x} \pm 1.96 \times s$
- 95% CI for μ: $\bar{x} \pm 1.96 \times \frac{s}{\sqrt{n}}$



Example

The Scenario: E-Commerce Delivery Times
An online retailer wants to estimate the true **average delivery time (μ)** for millions of packages shipped worldwide. They can't check every package, so they take a random sample.
Sample size (n) = 100 packages
Sample mean ($\bar{x}$) = 4.2 days
Population standard deviation (σ) = 0.8 days (known from historical data)

Find a **95% confidence interval** to estimate where the true population mean (μ) actually falls.
How about 85%, 80%?

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
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-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189

The Unavailability of σ

The first pivot needs σ. Nobody has σ. Replacing it by s breaks the pivot's distribution, not its structure.

A researcher in possession of σ would in almost all circumstances also possess μ, and no estimation would be needed. Substituting s puts a second random quantity into the denominator: the pivot becomes a ratio of two random things rather than one random thing over a constant. Its distribution is therefore no longer N(0,1), and the question is what it has become.

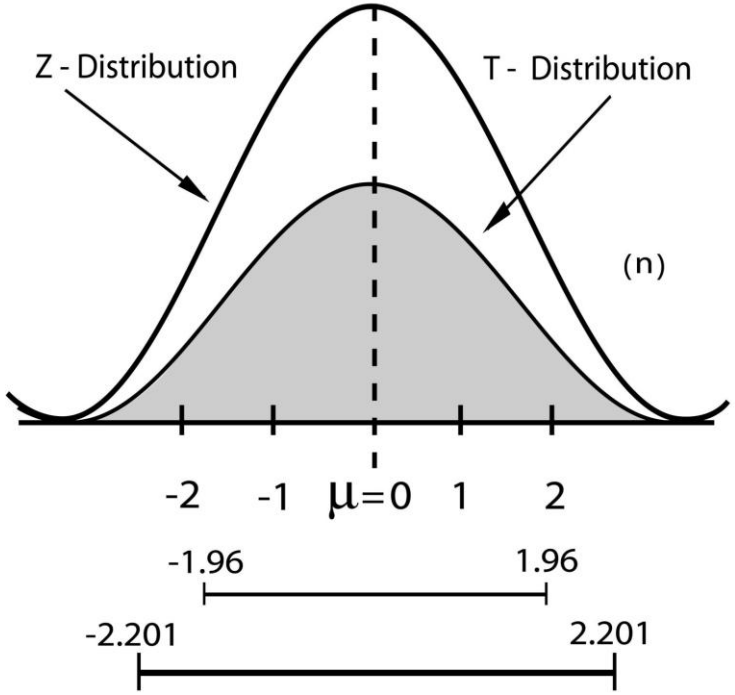
The substitution is unavoidable, and it is not free. The price is measurable.

Student's t Distribution

THE PRICE, MEASURED

A **t-statistic** (or t-score) measures how many standard errors a sample mean is from a hypothesized population mean **when you do not know the true population standard deviation (*σ*)**. Lower in the middle, heavier in the tails, symmetric about zero. Heavier tails mean a larger quantile, a larger quantile means a wider interval

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$



Use t unless σ is known. The rule is asymmetric because the two errors are: t when z would do costs almost nothing; z when t was needed understates the width.

Degrees of Freedom

df = n − (number of parameters the data itself had to supply)

Quantity computed	Parameters estimated from the same data	df
s^2 from n observations	1 — the mean, replaced by $\bar{x}$	n − 1
t interval for μ	1	n − 1
χ^2 interval for σ^2	1	n − 1
simple regression	2 — intercept and slope	n − 2
multiple regression, k regressors	k + 1	n − k − 1

The deviations satisfy $\Sigma(x_i - \bar{x}) = 0$ has exactly one linear constraint, exactly one degree lost.

Why the Divisor Is n − 1

E[s²] = σ² holds with the divisor n − 1. With the divisor n, the expectation is $\left(\frac{n-1}{n}\right)\sigma^2$.

Dividing by n produces an estimator that is too small on average — biased downward, in the exact sense defined in Part 1, by a factor that vanishes only as n grows. The n − 1 is the divisor that makes the estimator unbiased, and it is the same n − 1 that counts the degrees of freedom. One constraint, one degree lost, one adjusted divisor.

$$\sum_{i=1}^{n}(x_i-\mu)^2=\sum_{i=1}^{n}(x_i-\bar{x})^2+n(\bar{x}-\mu)^2$$

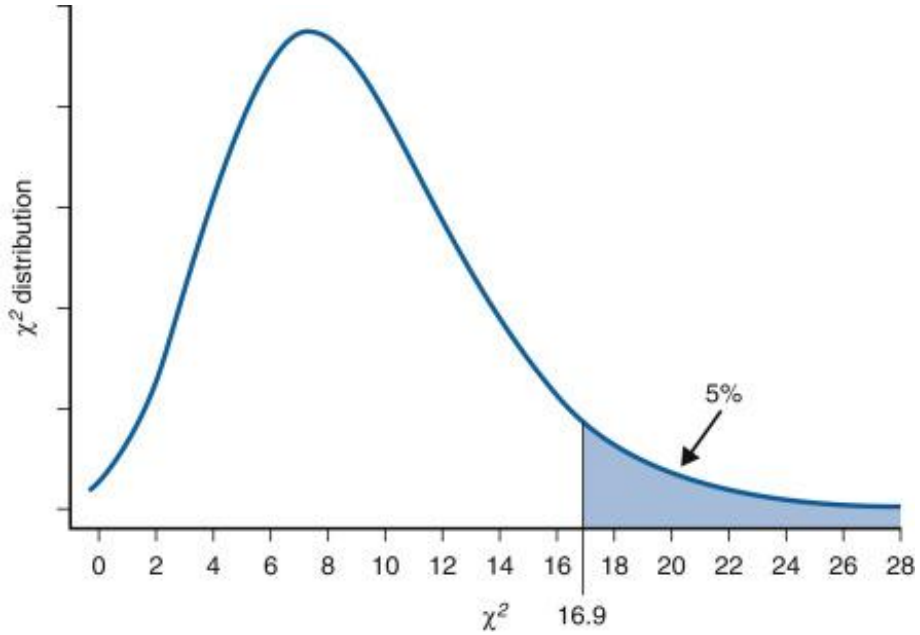
$$\mathbb{E}[\sum(x_i-\mu)^2]=\sum_{i=1}^{n}(x_i-\mu)^2=n\sigma^2$$

$$\mathbb{E}[n(\bar{x}-\mu)^2]\ =n(\bar{x}-\mu)^2=\ n\cdot \mathrm{Var}(\bar{x})=n\cdot\frac{\sigma^2}{n}=\sigma^2$$

The Chi-squared distribution

The Chi-squared (χ^2) distribution is a continuous probability distribution that is used primarily to analyze counts, categorical data, and variances.
Unlike the Z and T distributions—which track sample means—the Chi-squared distribution focuses on squared discrepancies between what you actually observe and what you mathematically expect to happen

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$



s^2 is non-negative and right-skewed, so no symmetric family could have described it.

The Chi-squared distribution

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

The chi-square family is the distribution of a sum of squared standard normal variables, and that is exactly what this expression is once the deviations are standardised — with one degree lost to the constraint $\Sigma(x_i - \bar{x}) = 0$.

$\alpha = 0.05$

$$P\left[\chi^2_{0.975} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{0.025}\right] = 0.95$$

$$P\left[\frac{(n-1)S^2}{\chi^2_{0.025}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{0.975}}\right] = 0.95$$

σ^2 appears in it, and its distribution does not depend on σ^2. Note the difference that matters: here the parameter sits in the DENOMINATOR.

Inverting the χ^2 Pivot

Step two, performed once more — and two consequences fall out of the algebra

Step	Result
The pivot, bracketed	$P(\chi^2(0.025) \le (n-1)s^2/\sigma^2 \le \chi^2(0.975)) = 0.95$
Solved for σ^2	$(n-1)s^2 / \chi^2(0.975) \le \sigma^2 \le (n-1)s^2 / \chi^2(0.025)$
Consequence 1	the LOWER bound uses the UPPER quantile — σ^2 was in a denominator, so the inequality reversed
Consequence 2	the interval is not symmetric about s^2, because the χ^2 family is not symmetric
In Excel	CHISQ.INV(0.025, n−1) and CHISQ.INV.RT(0.025, n−1)